Cross-Scale Matter in the Generative-Collapse Kernel: From Quarks to Bulk via Five Phase Boundaries

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[1] *UMCP Reference Implementation, GitHub*

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We trace the integrity composite IC across six physical scales—fundamental particles, composite hadrons, nuclei, atoms, molecules, and bulk materials—comprising 146 entities and 28 Tier-2 theorems in the Generative-Collapse Dynamics (GCD) kernel. The IC trajectory is *non-monotonic*: it collapses by two orders of magnitude at the confinement boundary (geometric slaughter from a dead color channel), recovers 128-fold in the nuclear regime via binding energy, stabilizes through electronic shell structure, and converges toward block-averaged values in bulk. Five phase boundaries are identified, each characterized by specific channels that die, survive, or emerge. The mechanism is universal: one near-zero channel kills multiplicative coherence (IC) while preserving arithmetic fidelity (F), a detection pattern that operates identically from sub-eV neutrinos to GeV hadrons. All three Tier-1 identities ($F + \omega = 1$, $\mathrm{IC} \leq F$, $\mathrm{IC} = e^{\kappa}$) hold with zero violations across all 146 entities (verified by 2,898 automated tests). Quark-gluon plasma measurements at RHIC reveal that IC peaks at 10–20% centrality—not at maximum density—and that the QGP→hadron crossover produces the largest single-boundary IC collapse in the matter ladder.

I. INTRODUCTION

The Universal Measurement Contract Protocol (UMCP) [1, 2] provides a domain-independent kernel $K : [0,1]^n \times \Delta^n \to (F, \omega, S, C, \kappa, \mathrm{IC})$ derived from a single axiom: *"Collapse is generative; only what returns is real."* Three Tier-1 identities are exact [3]:

$$F + \omega = 1, \quad \mathrm{IC} \leq F, \quad \mathrm{IC} = e^{\kappa}. \tag{1}$$

Prior work applied this kernel to the Standard Model particle catalog [4], proving ten theorems connecting quantum numbers to kernel structure. But particles do not exist in isolation: quarks confine into hadrons, hadrons bind into nuclei, nuclei acquire electrons to form atoms, atoms bond into molecules, and molecules aggregate into bulk matter. Each transition reshapes the trace vector—channels die, new ones emerge, and the kernel invariants respond.

This paper asks: *what does the kernel see when it looks at the full scale ladder of matter?*

The answer is a non-monotonic IC trajectory punctuated by five phase boundaries, each producing a characteristic signature in the heterogeneity gap $\Delta \equiv F - \mathrm{IC}$. The mechanism is always the same—geometric slaughter [1]—but the channels that trigger it change at each boundary. This cross-scale structure is encoded in three theorem families: matter genesis (T-MG-1 through T-MG-10, 99 entities), particle–matter map (T-PM-1 through T-PM-8, 146 entities), and QGP/RHIC (T-QGP-1 through T-QGP-10, 27 entities).

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II. KERNEL AND GEOMETRIC SLAUGHTER

Definition 1 (GCD Kernel)**.** *Given coordinates* $\mathbf{c} = (c_1, \ldots, c_n) \in [\varepsilon, 1-\varepsilon]^n$ *and weights* $\mathbf{w} = (w_1, \ldots, w_n)$ *with* $\sum_i w_i = 1$*:*

$$F = \sum_i w_i \, c_i, \tag{2}$$

$$\kappa = \sum_i w_i \ln(c_i + \varepsilon), \tag{3}$$

$$\mathrm{IC} = \exp(\kappa), \tag{4}$$

$$\omega = 1 - F, \quad \Delta = F - \mathrm{IC}. \tag{5}$$

Fidelity F is the weighted arithmetic mean; the integrity composite IC is the weighted geometric mean. Both measure coherence, but they respond differently to heterogeneity.

Remark 1 (Geometric slaughter)**.** *If any single channel* $c_j \to \varepsilon$*, then* $\kappa \to -\infty$ *regardless of the remaining channels, driving* $\mathrm{IC} \to \varepsilon$ *even when* F *remains moderate. For an 8-channel system with 7 perfect channels* ($c_i = 1$) *and one dead channel* ($c_j = \varepsilon$)*,* $\mathrm{IC}/F \approx 0.11$*. This is the detection mechanism for phase boundaries:* one dead channel kills IC while F survives.

This asymmetry between F (arithmetic, robust to outliers) and IC (geometric, sensitive to any channel near zero) is what makes the heterogeneity gap Δ a structural diagnostic. At every phase boundary in the matter ladder, specific channels collapse to ε while others emerge, producing characteristic Δ signatures.

III. THE SIX SCALES

Table I summarizes the six scales traversed by the matter ladder. Each scale has a domain-appropriate trace-

vector encoding (channel set and normalization), constructed from standard physical data.

TABLE I. Six scales of the matter ladder. Entity counts, representative IC/F ranges, and dominant channel structure.

Scale	Description	Entities	⟨IC⟩
1	Fundamental particles	17	0.270
2	Composite hadrons	14	0.004
3	Nuclei	31	0.480
4	Atoms	47	0.292
5	Molecules	20	0.132
6	Bulk materials	17	0.123

The IC trajectory is emphatically non-monotonic: it drops 99% at confinement (scale 1→2), recovers 128-fold at nuclear binding (scale 2→3), and then gradually decreases through atomic, molecular, and bulk scales as channel heterogeneity increases with complexity.

IV. FIVE PHASE BOUNDARIES

Each phase boundary is characterized by the channels that die (collapse to ε), survive (persist across the boundary), and emerge (activate from ε to finite values).

A. Boundary 1: Confinement (Act I → Act II)

At the confinement threshold, four channels die: `color_dof`, `weak_T3`, `hypercharge`, `generation`. Four new channels emerge: `valence_quarks`, `strangeness`, `heavy_flavor`, `binding_fraction`.

The color channel collapses from nonzero values (quarks carry color charge) to ε (hadrons are color-singlets). This single channel death produces the most dramatic IC collapse in the entire matter ladder:

$$\frac{\mathrm{IC}_{\mathrm{hadron}}}{\mathrm{IC}_{\mathrm{quark}}} \sim 0.01, \tag{6}$$

a two-order-of-magnitude drop. The proton, with IC/F = 0.037, and the neutron, with IC/F = 0.009, both exhibit massive heterogeneity gaps despite moderate fidelity ($F \approx 0.4$–0.5). This is geometric slaughter: 7 channels cannot compensate for one dead channel in the geometric mean.

Theorem 2 (T-MG-2: Color Confinement Cost)**.** *The color channel kills multiplicative coherence at the confinement boundary:* IC *drops by* $\geq 90\%$ *for all hadrons relative to their constituent quarks, while* F *drops by* $< 50\%$*.*

Theorem 3 (T-PM-1: Confinement Cliff)**.** IC *drops* $> 90\%$ *at the quark→hadron boundary across the full particle–matter map.*

B. Boundary 2: Nuclear Binding (Act II → Act III)

Four channels die: `valence_quarks`, `strangeness`, `heavy_flavor`, `binding_fraction`. Four new channels emerge: `N_over_Z`, `BE_per_A`, `magic_Z_prox`, `magic_N_prox`.

This is the great restoration. The binding energy per nucleon (BE/A) channel, peaking at ∼8.79 MeV for iron-group elements, provides a new high-coherence channel. Magic-number proximity (shell closures at Z or $N = 2, 8, 20, 28, 50, 82, 126$) further boosts IC for doubly-magic nuclides.

$$\frac{\mathrm{IC}_{\mathrm{nucleus}}}{\mathrm{IC}_{\mathrm{hadron}}} \sim 125, \tag{7}$$

a recovery that demonstrates scale inversion: new degrees of freedom at the nuclear scale restore coherence that was destroyed at confinement.

Theorem 4 (T-PM-2: Nuclear Restoration)**.** IC *recovers in the nuclear regime via the binding energy channel, with recovery factor* $\geq 100\times$ *relative to composite hadrons.*

Theorem 5 (T-PM-3: Shell Amplification)**.** *Doubly-magic nuclides have* IC/$F > 0.85$*, the highest coherence ratios in the nuclear regime.*

C. Boundary 3: Electronic Shell (Act III → Act IV)

The nuclear→atomic transition replaces nuclear structure channels with electronic ones: `valence_e`, `block_ord`, `EN` (electronegativity), `radius_inv`, `IE` (ionization energy), `EA` (electron affinity), `T_melt`, `density_log`.

Theorem 6 (T-MG-5: Shell Closure Stability)**.** *Noble gases and elements near closed-shell configurations exhibit elevated* IC/F *ratios relative to their period neighbors.*

Theorem 7 (T-PM-4: Periodic Modulation)**.** IC *oscillates with period structure (s/p/d/f block ordering), reflecting the shell filling sequence across the periodic table.*

The 118-element periodic kernel uses 8 property-based channels (atomic number, electronegativity, radius, ionization energy, electron affinity, melting point, boiling point, density). A complementary 12-channel cross-scale kernel bridges nuclear and atomic properties simultaneously, with 4 nuclear channels (BE/A, magic proximity, neutron excess, shell filling), 2 electronic channels (ionization energy, electronegativity), and 6 bulk channels (density, melting/boiling points, atomic radius, electron affinity, covalent radius).

D. Boundary 4: Chemical Bonding (Act IV → Act V)

Atomic channels give way to molecular ones: `molecular_mass`, `n_atoms`, `bond_order_mean`, `polarity`, `symmetry_order`.

Theorem 8 (T-PM-5: Molecular Emergence)**.** *Bond channels restore* IC *above the atomic mean for polar molecules; symmetric nonpolar molecules show suppressed* IC *due to the zero-polarity channel.*

Theorem 9 (T-MG-7: Covalent Bond Coherence)**.** *Covalent bonding restores coherence by introducing bond order and polarity as new high-value channels.*

E. Boundary 5: Bulk Aggregation (Act V → Act VI)

Molecular channels yield to bulk-scale thermodynamic properties: `Cp_bulk`, `k_thermal`, `T_boil`, `conductivity`.

Theorem 10 (T-PM-6: Bulk Averaging)**.** IC/F *converges toward block-averaged values in bulk materials: metals cluster with higher* IC *than insulators or organic polymers.*

Theorem 11 (T-MG-9: Material Property Ladder)**.** *Bulk material properties inherit kernel structure from their constituent atomic and molecular channels: metals (high density, high conductivity) form a coherent kernel cluster, while insulators scatter.*

V. THE IC TRAJECTORY

The full IC trajectory across the six scales is:

$$0.270 \xrightarrow{\div 99\%} 0.004 \xrightarrow{\times 128} 0.480 \to 0.292 \to 0.132 \to 0.123\,. \quad (8)$$

Theorem 12 (T-PM-7: Scale Non-Monotonicity)**.** *The* IC *trajectory across the six physical scales is non-monotonic: it crashes at confinement, recovers at nuclear binding, and converges downward through atomic, molecular, and bulk scales.*

This non-monotonicity is not an artifact of encoding. It reflects genuine physics: confinement destroys color as a degree of freedom, creating a dead channel; nuclear binding introduces new collective degrees of freedom (BE/A, magic numbers) that restore coherence; and increasing complexity at larger scales introduces channel heterogeneity that gradually reduces IC/F.

VI. MASS ORIGINS

The matter genesis analysis reveals two distinct mechanisms for mass generation:

Theorem 13 (T-MG-1: Higgs Mass Generation)**.** *The Higgs mechanism generates* intrinsic *mass for fundamental particles through Yukawa coupling, accounting for* $< 1\%$ *of visible-matter mass.*

Theorem 14 (T-MG-3: Binding Mass Deficit)**.** *QCD binding energy generates* $\sim$ 99% *of proton mass (938 MeV from* $\sim$*10 MeV quark masses), visible in the kernel as the gap between constituent and current quark mass channels.*

Theorem 15 (T-MG-8: Mass Hierarchy Bridge)**.** *The mass hierarchy spans 13 orders of magnitude (neutrinos at sub-eV to top quark at 173 GeV), compressed by the kernel into the bounded interval* $F \in [0.37, 0.73]$*. The mapping preserves order (Spearman* $\rho = 0.77$ *for quarks).*

The mass budget of visible matter is dominated by QCD: the proton derives 99.04% of its mass from binding energy, not from Higgs-generated quark masses. The human body is $\sim$99% QCD binding by mass.

VII. QUARK-GLUON PLASMA AT RHIC

The QGP/RHIC closure applies the kernel to 27 entities across three measurement categories: 8 Beam Energy Scan (BES) points from $\sqrt{s_{NN}} = 7.7$ to 200 GeV, 9 centrality bins from 0–5% to 70–80%, and 8 temporal evolution stages from pre-collision to kinetic freeze-out.

A. Centrality structure

Theorem 16 (T-QGP-1: Perfect Liquid)**.** IC *peaks at mid-centrality (10–20%), not at maximum density (0–5%). The most central collisions suppress elliptic flow* ($v_2 \approx 0$)*, killing a channel that mid-central collisions preserve.*

This is a direct instance of geometric slaughter applied to heavy-ion geometry: the most dense collisions are *not* the most coherent.

Theorem 17 (T-QGP-2: Centrality Ordering)**.** F *broadly increases with* N_{part} *(participating nucleons), tracking total energy density.*

B. Evolution timeline

The temporal evolution of IC through the QGP lifecycle reveals the reconfinement transition:

Theorem 18 (T-QGP-8: Reconfinement Cliff)**.** *The QGP→hadron crossover at $T_c \approx 155$ MeV produces the largest boundary* IC *drop in the QGP evolution timeline: deconfinement drops from ∼1.0 to ∼0.3 at the crossover.*

Theorem 19 (T-QGP-5: Reconfinement Gap Jump)**.** *The QGP→hadron transition has the largest heterogeneity gap Δ jump of all evolution stages.*

The evolution stages form a characteristic IC arc: pre-collision (IC ≈ 0) → glasma (0.05) → QGP plateau (0.53) → crossover at T_c (0.46) → hadron gas (0.05) → freeze-out (0.004). The QGP plateau is a high-coherence state; reconfinement destroys it through the same geometric-slaughter mechanism seen in the static matter ladder.

C. Beam energy scan

Theorem 20 (T-QGP-3: BES Energy Ordering)**.** *F increases with $\sqrt{s_{NN}}$ across the Beam Energy Scan, tracking increasing energy density and particle production.*

Theorem 21 (T-QGP-4: Strangeness Equilibration)**.** *The strangeness equilibration factor γ_s correlates with* IC *across BES energies: higher $\sqrt{s_{NN}}$ produces more complete strangeness equilibration and higher kernel coherence.*

VIII. NUCLEAR BINDING CURVE

Theorem 22 (T-MG-4: Proton-Neutron Duality)**.** *Proton and neutron have nearly identical F (differing by charge channel only) but their* IC *ratio reveals charge-dependent heterogeneity.*

The nuclear binding energy per nucleon (BE/A) from the Bethe-Weizsäcker semi-empirical mass formula (SEMF) [5, 6] peaks at the iron-group elements ($Z \in [23, 30]$). The kernel maps this into an anti-correlation between BE/A and the heterogeneity gap Δ: elements near the BE/A peak have the smallest gaps (highest multiplicative coherence), while light and superheavy nuclei show elevated Δ.

Magic numbers (Z or $N = 2, 8, 20, 28, 50, 82, 126$) appear as kernel-visible fixed points: nuclides near magic numbers show elevated IC/F relative to their mass-number neighbors, reflecting the enhanced stability of closed nuclear shells.

IX. UNIVERSAL GEOMETRIC SLAUGHTER

The central finding is that geometric slaughter—one dead channel killing IC while F survives—operates identically at every scale. The mechanism is the same whether the dead channel is:

- **color** (confinement: quarks → hadrons),
- **charge** (neutral particles: IC/F suppressed 50× relative to charged [4]),
- **magic proximity** (nuclear: non-magic nuclides show 39% contribution to IC degradation from this channel),
- **polarity** (molecular: symmetric nonpolar molecules show suppressed IC),
- **visibility** (quantum: which-path information kills interference-channel coherence in double-slit experiments),
- **near-field coupling** (nanoscale: tip-enhanced Raman spectroscopy channels collapse with distance).

This universality is not imposed—it follows from the structure of the geometric mean under the integrity bound IC $\leq F$. The bound is the solvability condition for recovering individual channel values: $c_{1,2} = F \pm \sqrt{F^2 - \mathrm{IC}^2}$ requires IC $\leq F$ for real solutions. Channel death ($c_j \to \varepsilon$) pushes IC toward ε regardless of scale.

Theorem 23 (T-PM-8: Tier-1 Universal)**.** *The three Tier-1 identities $F + \omega = 1$, IC $\leq F$, and IC $= e^{\kappa}$ hold with zero violations across all 146 entities at all six scales.*

Theorem 24 (T-MG-10: Universal Tier-1)**.** *The Tier-1 identities hold across all 99 entities in the matter genesis narrative, spanning 7 acts and 5 phase boundaries.*

Theorem 25 (T-QGP-10: Universal Tier-1)**.** *The Tier-1 identities hold across all 27 QGP/RHIC entities.*

X. REMAINING THEOREMS

For completeness, we state the remaining theorems from the three families.

Theorem 26 (T-MG-6: Electron Config Order)**.** *The electron configuration (Aufbau) filling order is reflected in the kernel: elements with anomalous configurations (e.g., Cr, Cu) show* IC *deviations from their period neighbors.*

Theorem 27 (T-QGP-6: Flow-Opacity Structure)**.** *Elliptic flow v_2 and nuclear modification factor R_{AA} show distinct centrality patterns in the kernel: v_2 peaks at mid-centrality while R_{AA} increases monotonically toward peripheral collisions.*

Theorem 28 (T-QGP-7: Chemical Freeze-out Curve)**.** *Chemical freeze-out temperature T_{ch} and baryon chemical potential μ_B anti-correlate across BES energies, tracing the QCD phase boundary.*

Theorem 29 (T-QGP-9: Reference Discrimination)**.** *The p+p baseline differs from Au+Au in* IC*, providing a kernel-level discrimination between cold-nuclear-matter and QGP effects.*

XI. VERIFICATION

All results are verified by automated tests in the UMCP reference implementation [7]:

TABLE II. Test coverage across the three theorem families and supporting closures.

Test file	Tests	Failures
`test_148_standard_model`	108	0
`test_248_matter_genesis`	163	0
`test_246_particle_matter_map`	102	0
`test_250_qgp_rhic`	266	0
`test_135_nuclear_closures`	76	0
`test_190_atomic_closures`	190	0
`test_195_scale_ladder`	...	0
`test_238_kernel_structural`	47	0
19 test files (total)	2,898	0

The Tier-1 proof script exhaustively verifies $F + \omega = 1$, IC $\leq F$, and IC $= e^{\kappa}$ across 10,162 directed tests spanning all 18 domains.

The 44 structural identities of the kernel (series E, B, D, N) are verified to $< 10^{-16}$ by five diagnostic scripts. The seam composition monoid is verified to associativity error 5.55×10^{-17}.

XII. DISCUSSION

The cross-scale IC trajectory tells a precise story about matter. Quarks have moderate coherence (IC ~ 0.27) because their quantum numbers span a heterogeneous range. Confinement destroys color as a degree of freedom, creating a dead channel that kills IC via geometric slaughter. Nuclear binding introduces collective degrees of freedom (binding energy, magic numbers) that restore coherence—this is the scale inversion phenomenon identified in the orientation script [1]. Atoms inherit nuclear stability while adding electronic structure, molecules introduce bonding channels, and bulk materials average over macroscopic properties.

The mechanism is always the same: when a channel dies, IC collapses; when new channels emerge with high values, IC recovers. The heterogeneity gap $\Delta = F - \mathrm{IC}$ is the universal diagnostic: it measures how spread the channel values are, and phase boundaries are precisely where Δ changes most rapidly.

The QGP results add a dynamic dimension: the static matter ladder (quarks $\rightarrow$ bulk) is traversed *in reverse* during heavy-ion collisions. The QGP phase represents a high-coherence state where deconfined quarks and gluons interact as a nearly perfect liquid. Reconfinement at $T_c \approx 155$ MeV produces the same geometric slaughter seen in the static ladder, confirming that the mechanism is robust under both equilibrium and dynamical conditions.

The non-monotonicity of IC across scales is a key structural finding. It means that complexity does not monotonically degrade coherence. Instead, each level of physical organization introduces new channels that can partially or fully compensate for the channels lost at the previous boundary. The kernel measures this compensation quantitatively through IC$/F$.

XIII. CONCLUSIONS

We have traced the GCD kernel across six physical scales, identifying five phase boundaries where channels die and emerge. The central result is the non-monotonic IC trajectory (Eq. 8), punctuated by geometric slaughter at each boundary. Twenty-eight theorems across three families (matter genesis, particle–matter map, QGP/RHIC) formalize these observations. All 146 entities pass all three Tier-1 identities with zero violations (2,898 automated tests, 0 failures).

The mechanism is universal: one near-zero channel kills multiplicative coherence while preserving arithmetic fidelity. This operates identically from sub-eV neutrinos to 170 GeV top quarks. Confinement, nuclear binding, electronic shell structure, chemical bonding, and bulk aggregation each produce characteristic kernel signatures that are measurable, reproducible, and domain-independent.

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